

A Minimal Copper-Plate Storage-Scheduling Benchmark for Differential Dynamic Programming

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I. COPPER-PLATE STORAGE-SCHEDULING MODEL

A. Optimization Problem

A single storage device and a single grid connection serve a known demand p_L^t over T periods of length Δt , all quantities in per unit. Following the sign convention of [1], $P_B^t > 0$ discharges and $P_B^t < 0$ charges; B^t is the energy at the *start* of period t , so B^1 is given and B^{T+1} is terminal. With $t \in \mathcal{T} = \{1, \dots, T\}$:

$$\min_{P_{\text{Subs}}, P_B, B} \sum_{t=1}^T \left[c^t P_{\text{Subs}}^t + C_B (P_B^t)^2 \right] \Delta t + w (B^{T+1} - B^1)^2 \quad (1a)$$

$$\text{s.t. } P_{\text{Subs}}^t + P_B^t - p_L^t = 0, \quad t \in \mathcal{T} \quad (1b)$$

$$B^1 = B_0 \quad (1c)$$

$$B^{t+1} = B^t - P_B^t \Delta t, \quad t \in \mathcal{T} \quad (1d)$$

$$P_{\text{Subs}}^t \geq 0, \quad t \in \mathcal{T} \quad (1e)$$

$$P_B^{\min} \leq P_B^t \leq P_B^{\max}, \quad t \in \mathcal{T} \quad (1f)$$

$$B^{\min} \leq B^{t+1} \leq B^{\max}, \quad t \in \mathcal{T} \quad (1g)$$

The *base instance* is (1a)–(1d), with all three bounds dropped.

B. Optimal-Control Form

Restated over $N = T$ stages in the form required by DDP:

$$\begin{aligned} \min_{x, u} \quad & \sum_{t=1}^{N-1} \ell(x^t, u^t) + \ell^F(x^N, u^N) \\ \text{s.t.} \quad & x^1 = \hat{x}^1, \\ & x^{t+1} = f(x^t, u^t), \quad t \in \{1, \dots, N-1\}, \\ & c(x^t, u^t) = 0, \quad t \in \{1, \dots, N\}. \end{aligned} \quad (2)$$

with state, control and initial condition

$$x^t = (B^t, \tau^t)^\top \in \mathbb{R}^2, \quad u^t = (P_{\text{Subs}}^t, P_B^t)^\top \in \mathbb{R}^2, \quad (3)$$

$\hat{x}^1 = (B_0, 1)^\top$. Period-varying data is carried by the time index τ through the Lagrange interpolants on the nodes $\tau = 1, \dots, T$,

$$p_L(\tau) = \sum_{k=1}^T p_L^k \prod_{\substack{m=1 \\ m \neq k}}^T \frac{\tau - m}{k - m}, \quad (4)$$

and likewise for the price $c(\tau)$. The stage data is then

$$f(x, u) = (x_1 - u_2 \Delta t, x_2 + 1)^\top, \quad (5)$$

$$c(x, u) = u_1 + u_2 - p_L(x_2), \quad (6)$$

$$\ell(x, u) = [c(x_2) u_1 + C_B u_2^2] \Delta t, \quad (7)$$

$$\begin{aligned} \ell^F(x, u) = & [c^T u_1 + C_B u_2^2] \Delta t \\ & + w (x_1 - u_2 \Delta t - B_0)^2. \end{aligned} \quad (8)$$

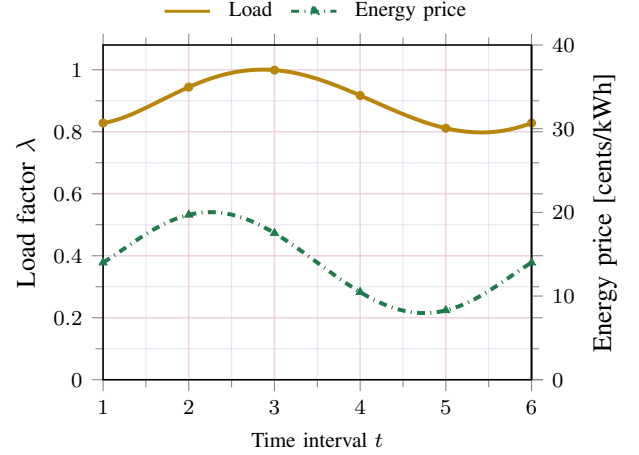


Fig. 1. Input profiles at $T = 6$: load factor (left axis), energy price (right). Markers are the data; the curves are the Lagrange interpolants, which is what the solver actually evaluates. Overshoot between nodes is slight (1.0005 against a node maximum of 0.999), but the interpolant degree is $T - 1$.

Bounds (1e)–(1f) map onto control limits $b^L \leq u^t \leq b^U$. The state bound (1g) is appended as a second row of c via a slack s^t adjoined to the control, giving $n_c = 2$, $n_u = 3$:

$$(x_1 - u_2 \Delta t) - B^{\min} - s^t = 0, \quad 0 \leq s^t \leq B^{\max} - B^{\min}. \quad (9)$$

C. Closed-Form Reference

The base instance is a strictly convex equality-constrained QP. Eliminating $P_{\text{Subs}}^t = p_L^t - P_B^t$ and $B^{T+1} - B^1 = -\Delta t \sum_t P_B^t$ reduces it to

$$\begin{aligned} J(P_B) = & \text{const} - \Delta t \sum_t c^t P_B^t + C_B \Delta t \sum_t (P_B^t)^2 \\ & + w (\Delta t)^2 s^2, \end{aligned} \quad (10)$$

with $s = \sum_t P_B^t$. Stationarity divided through by Δt is $-c^k + 2C_B P_B^k + 2w \Delta t s = 0$; summing over k collapses it to

$$s = \frac{\sum_t c^t}{2C_B + 2Tw \Delta t}, \quad P_B^k = \frac{c^k - 2w \Delta t s}{2C_B}. \quad (11)$$

REFERENCES

- [1] A. R. Jha, S. Paul, and A. Dubey, "Analyzing the performance of linear and nonlinear multi-period optimal power flow models for active distribution networks," in *2025 IEEE North-East India International Energy Conversion Conference and Exhibition (NE-IECCCE)*. IEEE, 2025, pp. 04–06.

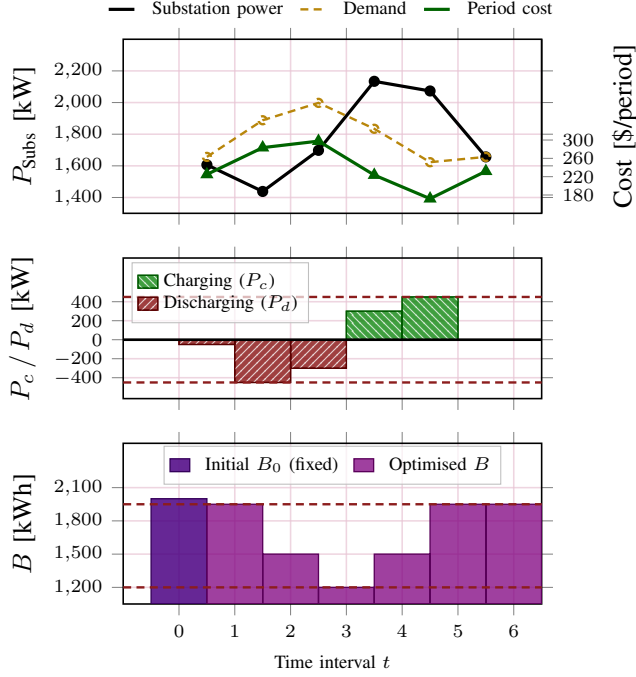


Fig. 2. The benchmark at $T = 6$ with all bounds imposed. Charging is drawn upward, discharging downward, so a bar's direction is the device's action rather than the sign of P_B^t . The battery discharges through $t = 1-3$ and recharges at $t = 4, 5$, riding its ± 450 kW rails and both energy limits (dashed). Those five bounds are the active set the reference solver reports.

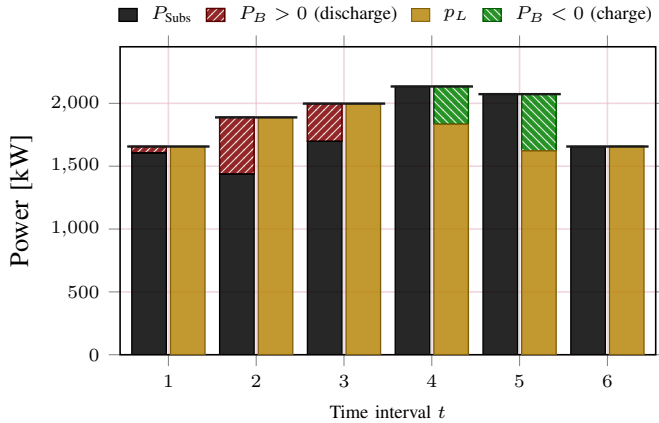


Fig. 3. Power-balance composition at $T = 6$. Per interval, supply (P_{Subs}^t plus discharge) on the left against demand (p_L^t plus charge) on the right; the equal heights are the power balance, marked by the rule. At $t = 2$ the battery supplies 23.8% of the node's 1888 kW; at $t = 5$ its charging instead raises import to 2073 kW to serve 1623 kW of load.

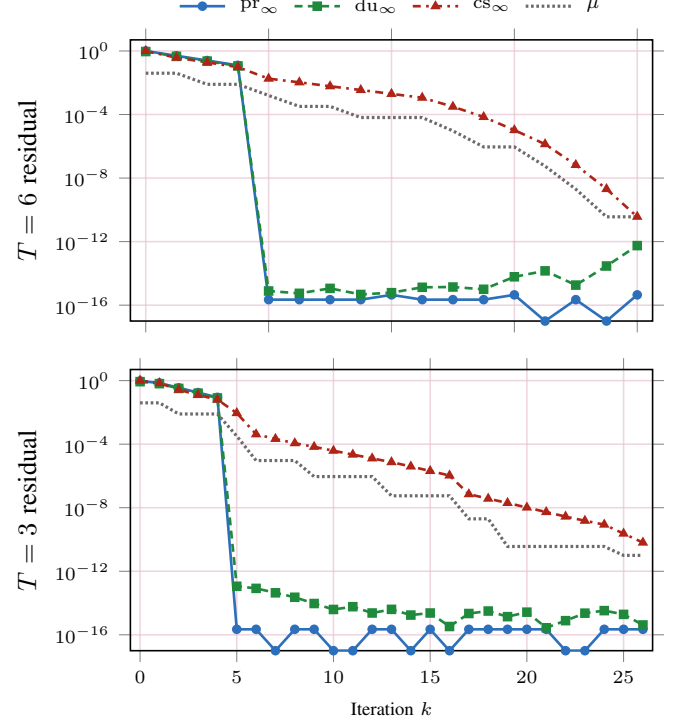


Fig. 4. FilterDDP convergence, all bounds imposed: $T = 6$ (top, 17 iterations) and $T = 3$ (bottom, 27). Feasibility is reached almost immediately — both pr_∞ and du_∞ hit machine precision by $k = 4$ — after which progress is entirely the barrier parameter μ driving cs_∞ down alongside it. Iteration count here is set by the μ schedule, not by the optimisation.

TABLE I
FILTERDDP ITERATIONS, $T = 6$, ALL BOUNDS

k	objective	backward pass				forward pass	
		pr_∞	du_∞	cs_∞	$\log_{10} \mu$	α	bk
0	0.840000	1.0e+00	9.2e-01	1.0e+00	-1.40	0.50	1
1	1.225783	5.0e-01	4.6e-01	3.8e-01	-1.40	0.50	1
2	1.397002	2.5e-01	2.3e-01	1.9e-01	-2.10	0.50	1
3	1.464625	1.2e-01	1.1e-01	9.6e-02	-2.10	1.00	0
4	1.520935	2.2e-16	7.8e-16	1.8e-02	-2.80	0.50	1
5	1.507179	2.2e-16	5.6e-16	1.1e-02	-3.49	0.50	1
6	1.496237	2.2e-16	1.1e-15	6.0e-03	-3.49	0.50	1
7	1.488496	2.2e-16	4.7e-16	3.5e-03	-4.19	0.50	1
8	1.483187	4.4e-16	6.1e-16	2.0e-03	-4.19	0.50	1
9	1.479927	2.2e-16	1.3e-15	1.1e-03	-4.19	1.00	0
10	1.476349	2.2e-16	1.4e-15	3.1e-04	-5.03	1.00	0
11	1.475395	2.2e-16	1.0e-15	7.0e-05	-6.04	1.00	0
12	1.475195	4.4e-16	6.1e-15	1.0e-05	-6.04	1.00	0
13	1.475179	0.0e+00	1.4e-14	1.4e-06	-7.25	1.00	0
14	1.475173	2.2e-16	1.8e-15	6.7e-08	-8.70	1.00	0
15	1.475173	0.0e+00	2.9e-14	2.0e-09	-10.44	1.00	0
16	1.475173	4.4e-16	5.6e-13	3.7e-11	-10.44	—	—

The backward pass evaluates the residuals at the current iterate; the forward pass is the filter line search that follows it, with accepted step α and $\text{bk} = \log_2(1/\alpha)$ backtracks. FilterDDP prints these two halves one iteration out of step, because `iteration_status` is called before `forward_pass!`; the rows here are realigned so each holds one backward pass and the forward pass it actually fed. The last row has no forward pass — the solver converged instead of stepping. Regularisation never fired. Every backtrack in the table was a rollout rejection rather than a filter or sufficient-decrease rejection, and the solver's own `ls` counter records none of them.

TABLE II
INSTANCE DATA

Interval t	1	2	3	4	5	6
<i>Horizon $T = 6$</i>						
Demand p_L^t	1.657	1.888	1.998	1.834	1.623	1.657
Energy price c^t	0.140	0.197	0.175	0.105	0.083	0.140
<i>Horizon $T = 3$ — resampled, not a prefix of the above</i>						
Demand p_L^t	1.657	1.943	1.657	—		
Energy price c^t	0.140	0.140	0.140	—		
Period length Δt	1	Initial energy B_0		2.0		
Throughput C_B	0.05	Terminal weight w		5.0		
<i>Bounds, constrained variants only</i>			$T = 3$		$T = 6$	
$P_B^{\min}, P_B^{\max}$			−0.5, 0.004		−0.45, 0.45	
$B^{\min}, B^{\max}$			1.9895, 1.9965		1.20, 1.95	

All quantities are in per unit on the tADMM base, 1 p.u. = 1000 kW and 1 p.u.h = 1000 kWh; with $\Delta t = 1$, power and energy are numerically interchangeable. Figs. 2 and 3 report battery quantities in kW and kWh rather than per unit. At $T = 6$ the bounds below are ± 450 kW and 1200–1950 kWh. The series are the tADMM load and price profiles, $p_L^t = 2\lambda^t$ with $\lambda^t = 0.8 + 0.1(\sin(\theta^t - 0.8) + 1)$ and $c^t = 0.08 + 0.06(\sin \theta^t + 1)$, sampled at $\theta^t \in \text{range}(0, 2\pi, T)$. They therefore *resample* with T : the $T = 3$ row is not the first three entries of the $T = 6$ row. Two consequences of that sampling are worth stating plainly. First, both endpoints of $[0, 2\pi]$ are included, so $c^1 = c^T$ for every T . Second, at $T = 3$ the samples fall at $0, \pi$ and 2π , where $\sin$ vanishes, so *the price is constant* and the $T = 3$ instance carries no arbitrage signal at all — its optimal P_B^t is the same in every period and is set purely by the terminal penalty. The -0.8 phase offset on the load is what keeps the two series from being collinear; their Pearson correlation is 0.644 at $T = 6$. Bounds apply only to the constrained variants. Figs. 1, 2 and 3 all show the $T = 6$ case.